London Petro Academy Ltd

Renewable Energy Economics & Finance

Evaluating renewable technologies, modelling project economics and structuring bankable finance for sustainable power assets.

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Empowering Professionals for a Sustainable Future

2026/2027

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Course Overview

This multi-day Renewable Energy Economics & Finance course provides a practical, end-to-end understanding of how renewable power projects create value, manage risk and secure funding. It begins with the economics of key renewable technologies—wind, solar PV, hydro, biomass and geothermal—covering resource characteristics, capacity factors, cost structures and levelised cost of electricity across different locations. Participants then explore energy/resource uncertainty and bankability, learning how measurement campaigns, resource assessments and p50/p90 analysis affect debt capacity and project returns. Building on this, the course introduces project finance concepts and structures for renewables, including corporate vs project finance, stakeholder roles, risk phases, EPC and O&M contract requirements, and the tools used to manage technical, commercial and political risks. Finally, delegates work with loan term sheets, cash-flow waterfalls and financial models to calculate DSCR and coverage ratios, structure repayment profiles and evaluate bankable PPAs and tariff designs using case studies and modelling tools such as SAM, RETScreen and Excel.

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COURSE OBJECTIVES

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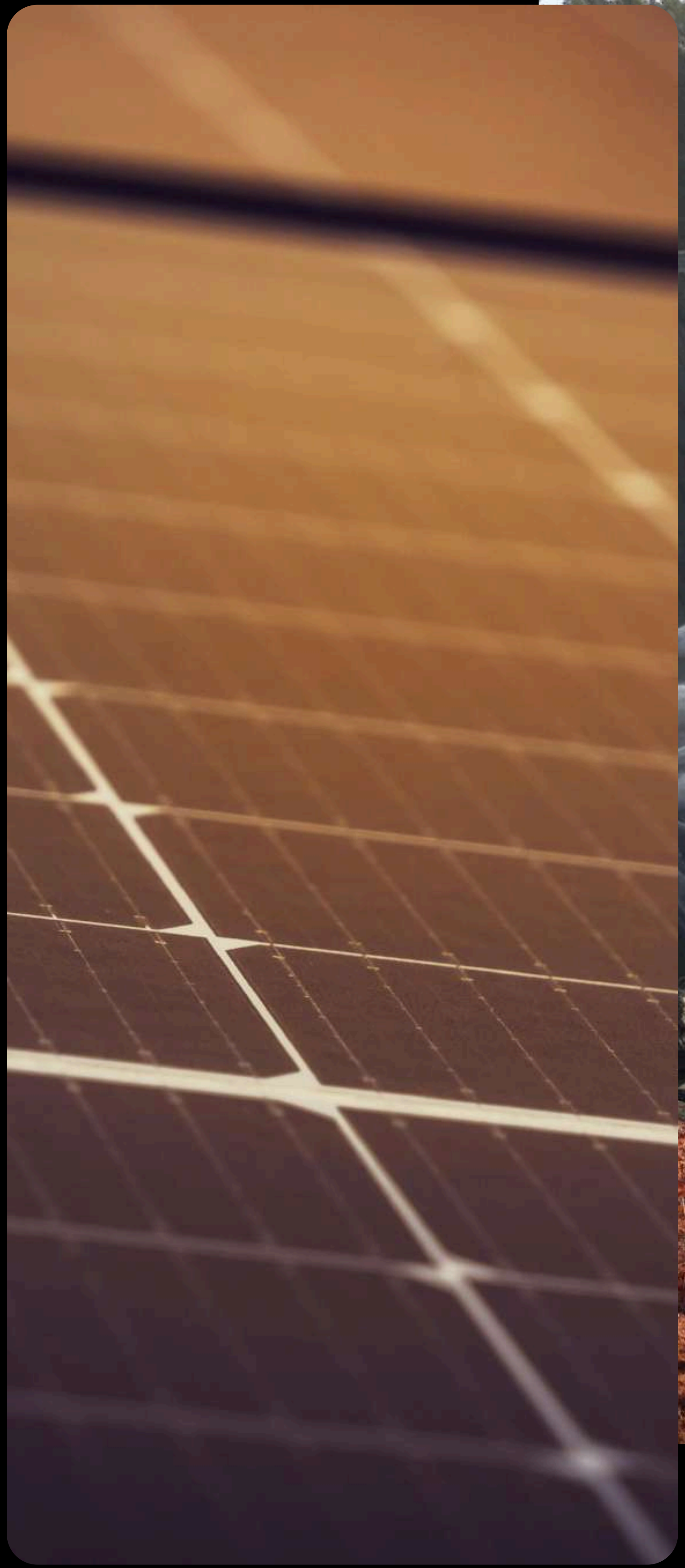
By the end of this course, participants will:

- Understand and compare the economics of major renewable energy technologies, including key cost drivers, performance metrics and resource considerations.
- Grasp the importance of resource assessment and energy uncertainty, and how reducing production uncertainty improves debt capacity and project bankability.
- Learn core concepts of renewable project finance, including capital structure, stakeholder roles, risk phases, and the role of EPC, O&M, warranties, insurance and hedging in risk mitigation.
- Interpret and use loan term sheets, cash-flow waterfalls and financial models to calculate CFADS, DSCR, LLCR/PLCR, NPV and IRR, and assess the impact of CAPEX, OPEX and revenue changes.
- Evaluate the requisites of a bankable PPA, different tariff structures and contract clauses (curtailment, deemed generation, payment terms) and understand their implications for lenders and investors.

These objectives are delivered through presentations, worked examples, group exercises, and real-world case studies from multiple technologies and markets.

Course audience

This course is designed for professionals who need a solid, commercially grounded understanding of renewable energy project economics and finance. It is particularly relevant for project developers and sponsors, OEMs, EPC contractors and O&M service providers, IPPs, utilities and grid operators involved in planning and delivering renewable assets. Private equity and institutional investors, development finance institutions, commercial lenders, and legal, technical and insurance advisors will benefit from the focus on bankability, risk allocation and financial modelling. Government officials, regulators, oil and gas upstream companies, petrochemical and process industry firms, corporate procurement and contract managers, and business developers and sales managers will also find the course valuable for understanding how renewable projects are evaluated, financed and de-risked in practice.



Economics of Renewable Technologies:

1

Wind

- Rotor blades as a key item for wind turbines defining power curve and energy production
- Drive trains and how their reliability impacts cost of energy and service costs
- Larger rotor diameters, increasing tower heights – is the sky the limit?
- Are the teething problems of offshore technologies overcome or why offshore at all?
- wind hours, wind speed, time of measurement, tower height, turbine performance

Solar Photovoltaic

- From silicon, ingots, wafer, cells to modules and solar PV power plants – limits of growth?
- Silicon, thin film, multijunction, organic and concentrated photovoltaics – where is the future?
- Balance of system including inverters, transformers, cabling and structural components
- From global irradiation to direct, diffuse and horizontal irradiation
- Watt-peak, cell efficiency, performance ratio and standard test conditions – Watt means what?
- One-axis and two axis tracking system – increased energy output or headache?

Hydro Power

- Catchment area, mean rain falls, mean flow
- Head availability, hydraulic structures (dike, dam, intake), run-o-river vs dam projects
- Turbine technologies
- Water quality, reserved flow, groundwater flow, sediment transport
- Flood prevention, fish passages, leisure activities

1

Biomass-Power

Co-firing versus dedicated biomass power plants for combined heat and power

From combustion steam cycle to Stirling engine, ORC, BICGT/BIGCC

Biogas via anaerobic digestion for small scale and off-grid installations

Difference between power generation efficiency and energy efficiency

Pelletisation, torrefaction, pyrolisis and gasification – where are they commercially?

Is it about technology or feedstock logistics and competition?

The bankability challenge of standalone biomass power plants

Importance of size and location for biomass power plants

Bankable feedstock supply for biomass power plants – quality, quantity, price and on time

Geothermal Power

Flash& Dry steam plants, Binary Kalina cycle(ORC) plants and their applications

Challenges in drilling and exploration – high upfront DEVEX or sunk costs?

Limits of scalability for geothermal power plants

Enhanced geothermal systems – leaving demonstration phase?

Challenges for heat usage and bankable revenues

drilling depth, # dwells, resulting water pressure & temperatures, initiated water cycle

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2

Energy Uncertainty & Resources Assessment

- Satellite and terrestrial data available for renewable resource assessment
- Measurement campaigns in order to acquire bankable annual energy production estimates
- Uncertainties of power curves or it wasn't like this in the brochure
- Understanding the resource assessment report - standard deviation, variance and p50/p90
- Experience of predictability and variability of renewable resources assessments over time
- Bankability impact of using one-year or ten year data and additional measurement investments
- How reducing uncertainties of annual energy output drives debt capacity and financial returns
- Sources of uncertainties and variability – expertise or layman understanding of the variables

Economics & Cost of Re Technologies

- Variables of capacity factors and availability and their differences
- CAPEX & OPEX split, variations and trends
- Understanding Levelized Cost of Electricity and why they are depending on location

RE finance and structures

- Why the financial concept should be integral part of the project development process?
- Key Differences of and rational for Corporate Finance and Project Finance
- Development of Project Finance within the Infrastructure and Renewable Sector
- Project financing structures and the complex commercial relationships and legal issues
- What means a bankable project and how to understand and meet lender's ratios
- Project Stakeholders – Developer, Sponsor, Lender, Insurance provider, EPC, Equipment Vendor, O&M, Authorities, Off-taker, Supplier
- Forming a robust project set-up from Equity and Debt Sources

2

Risk Management

- Risk phases: Development, Construction, Start-up and Operations & Maintenance
- Trigger Points: Concession signing, Financial Close, Commercial Operation Date, End of Loan
- EPC turn-key contract – on time, on budget, to specifications
- Delayed & Performance Liquidated Damages and Caps on Liability
- Force Majeure provisions and Dispute resolution
- Key issues on O&M service provider and contract in relation to bankability
- Warranties, insurances, guarantees, hedges – interest rate, currency, political risk
- Multilateral banks and export credit agencies – essential door openers for emerging markets

RE project screening checklists

- Project description and characteristics
- Land lease
- Resource Assessment
- Permits & Concessions
- Grid connection
- Project Contracts (EPC, PPA, Sponsors, Lenders, O&M, Insurance, Guarantees)
- Bankability, Due Diligence and external Advisors (technical, commercial, legal)
- Financial Analysis and Debt capacity

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Loan Term Sheet Elements

- Capital injections into the project, pro rata basis and timing of drawdowns
- Additional lenders security using Grace Period, Debt Service and O&M Reserve Accounts
- Cash Sweeps, Cash Traps and Dividend Blockers and how they damage Equity returns
- Where the various elements relate to within the Cash Flow Waterfall
- Covenants, Securities and the concept of Sources and Uses
- Interest Rate, Banking Fees, Margins and Commissions applied to a Project Finance Loan
- Conditions Precedents - Important steps towards Financial Close
- Direct Agreements

Financial Analysis & Structuring

- Using the Cash Flow Waterfall to calculate Cash Flow Available for Debt Service (CFADS)
- How the Cash Flow Prediction, the DSCR and loan tenor drives overall Debt Capacity
- Leveraging equity returns by gearing the equity:debt ratio
- Project and Equity NPV and IRR
- Limiting factors on loan tenors
- Difference between minimum and average DSCR over the loan life of the project
- Calculating Loan Life and Project Life Cover Ratios and comparing them to DSCR
- Different Repayment structures from linear, annuity and debt sculpting
- Sensitivity & Scenario Analysis applying to CAPEX, OPEX and Revenue variations

Power Purchase Agreements

- Off-take agreements for power, heat and certificates – importance for long-term floor
- Requisites of a bankable PPA
- PPA Timing Requirements and payment terms
- Different Tariff Structures for dispatchable and non-dispatchable technologies
- Capacity & Energy Charges, Metering and delivery point
- Cure Periods and Step-In Period – deemed commercial operation date
- Curtailment provisions within PPAs – deemed capacity and deemed generation clauses
- Risk of non-payment by the offtaker, FX volatility and currency convertibility risk

Case Studies:

- Case studies for the various technologies will show best practice, failures and the variety of applications for renewable
- Understanding resource audits and annual energy production estimates for various technologies and sites.
- Sample PPA, EPC and O&M contracts and clauses from various countries and technologies
- Sample Loan term sheets for renewable project finance
- SAM and RETScreen free available software for feasibility analysis of energy projects
- Excel based financial models to analyse LCOE and financial ratios

This course is available In-House and can be tailored to your specific requirements.

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Touchpoints

Question about this course?

Call us on +44 (0) 1582 516247 and one of our representatives will assist you with your enquiry.

3 Easy ways to register

Online: www.londonpetroacademy.co.uk

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